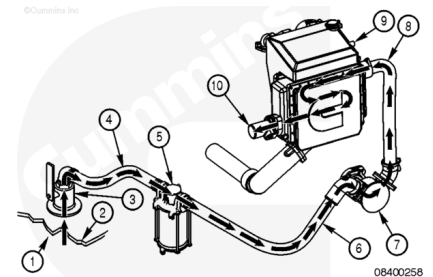


Trainings ▾

General Information

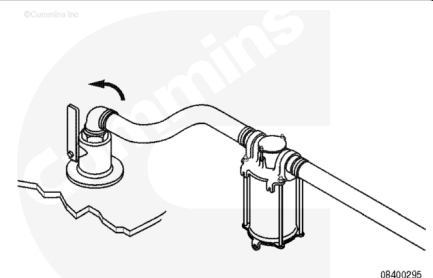
Typical sea water cooling system:

1. Grate-type strainer located on the bottom of the boat hull
2. Vessel bottom/hull
3. Sea water inlet valve
4. Hose
5. Sea water strainer
6. Sea water pump
7. Tube
8. Heat exchanger
9. Sea water outlet.



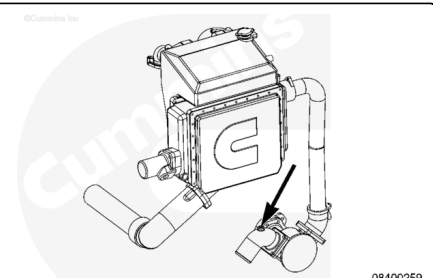
Inspect

Locate and verify that the sea water inlet valve is in the full-open position. If closed or partially closed, open the valve and check the engine temperature.

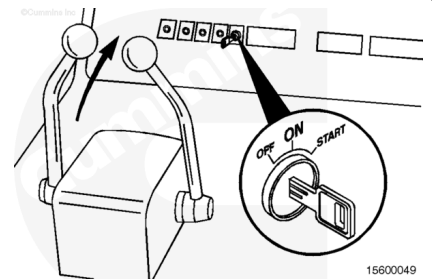


Test

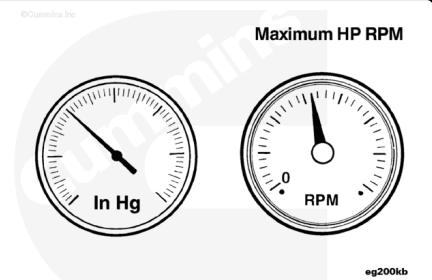
Attach a vacuum gauge to the inlet side of the sea water pump.



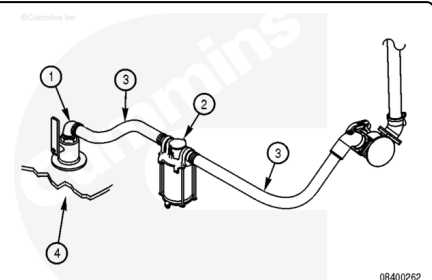
Start the engine.



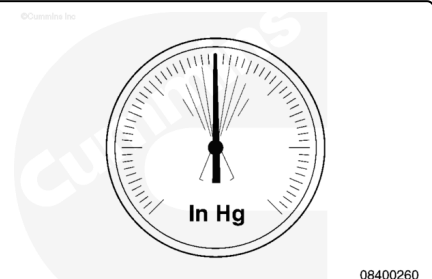
Check sea water restriction from idle to rated speed at 200-rpm increments while the vessel is in the dock. Sea water restrictions **must** be less than 17 kPa [5 in-Hg] under all operating conditions.



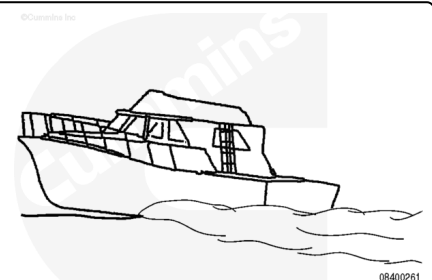
If the sea water inlet restriction is above specification, the source of the restriction (1,2,3 or 4) **must** be found. Troubleshooting and repair of excessive sea water restriction is a boat manufacturer or boat owner responsibility and **not** covered by Cummins® warranty.



If the sea water restriction readings fluctuate, inspect the sea water supply for leaks. These components are **not** built or supplied by Cummins Inc. Troubleshooting and repair of sea water aeration is **not** covered under Cummins® warranty.

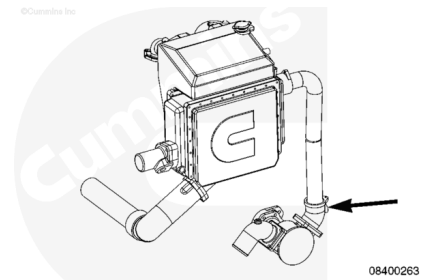


It is possible that aeration (gauge fluctuation) will **only** occur while the vessel is underway due to the introduction of air.

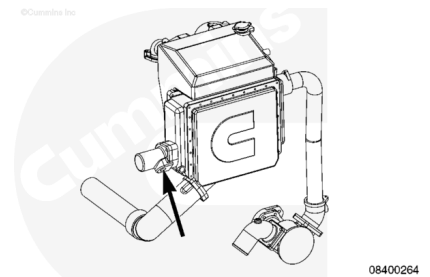


Pressure Test

Attach a pressure gauge to the outlet side of the sea water pump to check pump outlet pressure. Check water pump specifications for proper pressure. Record pressure reading for next step.



Attach the pressure gauge to the discharge side of the heat exchanger and check for pressure drop. Be certain to check both pressures under the same operating conditions- for instance, both at idle or rated speed.



If the pressure drop between the inlet side of the heat exchanger and the outlet side of the heat exchanger is greater than specification, check for blockage in the heat exchanger.

